Technical report of the final exam of the course "Statistical models for Healthcare data"

Prognostic factors and treatment efficacy in the Rotterdam breast cancer cohort: a multi-model analysis

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Abstract:
Background: Breast cancer is a heterogeneous disease and despite decades of research, substantial uncertainty remains regarding how individual prognostic factors combine to shape long-term outcomes, and how treatment effects can be reliably estimated in observational settings where therapies are not randomly assigned. This project investigates how individual clinical and pathological characteristics impact the probability of recurrence within six years from surgery and the probability of death within ten years from surgery. Furthermore, it evaluates the effectiveness of adjuvant therapies. **Methods**: data from 2,982 patients in the Rotterdam Breast Cancer Cohort were analyzed. Binary logistic regression models were developed to predict tumor recurrence and long-term mortality; time-to-event models (Cox and Kaplan-Meier) were used to estimate survival outcomes. Additionally, a causal inference analysis, using propensity score estimation and inverse probability of treatment weighting (IPTW), was conducted to calculate the effect of hormonal therapy on the outcomes. **Results**: lymph nodes involvement and histological grade emerged as critical risk factors, while smaller tumors had significant protective effects. Causal analysis shows that the hormonal treatment reduces the probability of mortality by 24% and tumor recurrence by 15%. **Conclusion:** this study confirms that lymph nodes involvement and histological grade are critical prognostic drivers in breast cancer. It demonstrates that standard predictive models can distort treatment efficacy due to confounding by indication. By applying causal inference frameworks, the protective benefits of adjuvant hormonal therapy are obtained, underscoring the necessity of these methods when evaluating treatments in observational healthcare settings.

1. Introduction

The prognosis of breast cancer depends on a complex combination of patient and tumor characteristics and treatment strategies. Clinical management decisions, such as the administration of adjuvant hormonal therapy or chemotherapy, are informed by risk stratification models that estimate the likelihood of recurrence and mortality. As suggested by Royston and Altman [1], these models must undergo rigorous external validation before entering clinical practice.
The efficacy of therapies in improving survival is demonstrated by a meta-analysis by the Early Breast Cancer Trialists' Collaborative Group, that revealed that 5 years of adjuvant hormonal therapy safely reduces 15-year risks of breast cancer recurrence and death in estrogen receptor-positive disease [2]. However, predicting the timing of recurrence remains a clinical challenge; some studies indicate that patients continue to face a significant risk of late recurrence even 10 to 32 years after primary diagnosis [3].
The aim of this study is to explore the impact of patient and tumor characteristics on recurrence and mortality, using the Rotterdam Breast Cancer Study cohort. The study was guided by some key research questions: (1) to what extent do baseline clinical and pathological factors influence recurrence and long-term mortality risk; (2) how does the implementation of feature selection and regularization methods impact the performance of the models compared to a full-feature approach; (3) what is the true Average Treatment Effect (ATE) of adjuvant hormonal therapy when selection bias is mitigated through IPTW and how does it differ from unadjusted observational estimate?
Based on prior findings [4], it is hypothesized that adverse baseline tumor characteristics (higher grade and positive nodes) will strongly predict higher recurrence and mortality risk. It is also expected that adjuvant hormonal therapy exhibits a significant causal protective effect on survival and recurrence when adjusting for measured confounders [2].

2. Materials and Methods

2.1 Dataset

The dataset comprises 2982 primary breast cancer patients whose records were included in the Rotterdam tumour bank. The data available are: patient identifier (pid), age at diagnosis (age), year of surgery (year), hospital facility (hospital_id), menopausal status (meno), number of positive lymph nodes (nodes), tumor size (size, ≤20, 20–50, and >50 mm), histological grade (grade), estrogen (er) and progesterone (pgr) receptor levels measured in fmol/l, hormonal treatment (hormon) and chemotherapy (chemo), time to recurrence (rtime) and time to death (dtime), with their event indicators (recur and death). During the data exploration it was observed that the distribution of the number of positive lymph nodes (nodes) exhibited a marked asymmetry, so, a logarithmic transformation was applied to the variable. Furthermore, hormone receptor levels (pgr and er) were binarized (pgr_pos, er_pos). In accordance with the conventions adopted in the study "Relevance of breast cancer hormone receptors and other factors to the efficacy of adjuvant tamoxifen" [1], was used a clinical threshold of 10 fmol/l: values exceeding this cut-off define the patient's receptor positivity. The histological grade was converted into a binary variable (grade_bin) to distinguish between low/intermediate and high tumour grades. Since the analysis intends to isolate the effect of patient and tumour characteristics, non-strictly clinical variables (pid, year, hospital_id) were excluded from the analysis.

2.2 Classification models to predict tumour recurrence and long-term mortality

In this first phase, the main objective is to develop a predictive model capable of estimating the probability of tumour recurrence within six years from surgery (recur) and one capable of estimating the probability of death within ten years from surgery (death). The research question underlying this section aims to determine to what extent baseline clinical and pathological characteristics can discriminate patients at a higher risk of adverse outcomes. The chosen timeframes capture the peak recurrence window, typically observed within 3 to 5 years [3], and aligns with the standard 5 years duration of adjuvant hormonal therapies like Tamoxifen [2]. Consequently, all patients who were censored within the first six years for recurrence and ten for death were excluded from the dataset. It was then divided into two parts: a training set, used for model train, and a test set, used for final evaluation. Given the need to estimate a dichotomous outcome (presence or absence of recurrence, death or not), the chosen method is based on binary logistic regression. To identify the most robust and best-performing model, three different modelling approaches (Full Model, Feature Selection Model and Regularized model) were developed and compared by analysing the area under the curve (AUC) obtained from the test set predictions. The optimal models were then trained on the entire training set and a 10-fold cross-validation was performed to ensure the robustness of the results and to verify the absence of overfitting. The final performance was evaluated using the confusion matrix.

2.3 Time-to-event models to examine recurrence-free and overall survival

The objective of this phase is to estimate and model recurrence-free and overall survival utilizing the entire follow-up period. The dataset was previously split into a training set for model development and a test set for final evaluation. The investigation was divided into two main phases:

- Non-parametric analysis: first, the Kaplan-Meier estimator was used to calculate survival probabilities over time and plot the corresponding curves. Furthermore, to compare survival trends among different clinical subgroups of interest (e.g., patients treated with hormonal therapy versus untreated patients, or stratifications by histological grade), the Log-Rank Test was applied.
- Semi-parametric analysis (Cox Model): this approach allowed for the estimation of the Hazard Ratios (HR) associated with each clinical predictor.

To identify the optimal predictive model, four configurations were compared on the training set: a baseline full model, a reduced model excluding non-significant predictors (meno, p_value=0.93, and er_pos, p_value=0.65), feature selection models utilizing forward and backward approaches and regularized models. The performance evaluation and the selection of the optimal model among the four proposed were based on the Concordance Index (C-index), calculated for each model through cross-validation procedures. Finally, before proceeding with the clinical interpretation of the Hazard Ratios on the definitive model, the validity of the fundamental proportional hazards' assumption was verified by analysing the Schoenfeld residuals.

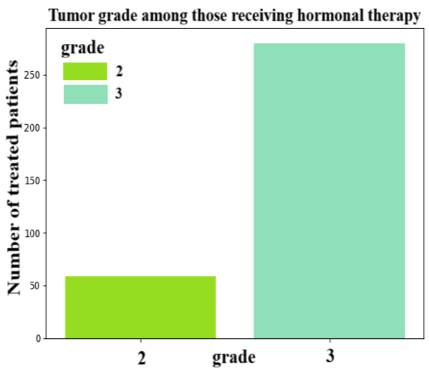


Figure 1. confounding by indication: who receive hormonal therapy has higher tumor grade

2.4 Causal inference analysis to estimate the effect of adjuvant hormonal therapy on recurrence and death

The final phase of the study was dedicated to causal inference, with the objective of estimating the direct effect of hormonal therapy (hormon) on the risk of tumour recurrence and death. To ensure the comparability of the results with the landmark meta-analysis on tamoxifen efficacy [2], the cohort was restricted to Estrogen Receptor-positive (ER+) patients. Since the study is based on observational data rather than a randomized clinical trial, the treatment assignment is not random but intrinsically linked to the patients' baseline clinical characteristics (confounding by indication, *Figure 1*). To recreate the conditions of a randomized experiment, as first thing was estimated the conditional probability for each patient of receiving hormonal therapy, given their baseline clinical characteristics (propensity score estimation). Given that the propensity score distribution for the untreated cohort was skewed to the left, an exploratory analysis was conducted to identify patient profiles with a zero probability of receiving treatment. The analysis revealed that patients with zero positive lymph nodes (log_nodes = 0) were never treated, while those who received chemotherapy (chemo = 1) accounted for only 9% of the overall treated population. Consequently, to strictly satisfy the positivity assumption (*Figure 2*), these specific subgroups were removed from the dataset. Finally, by applying the inverse probability of treatment weighting (IPTW) it was possible to calculate the Average Treatment Effect (ATE).

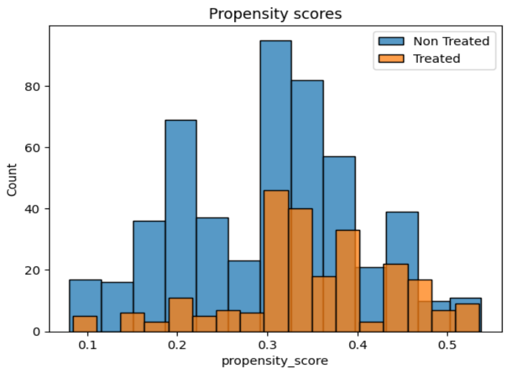


Figure 2. Overlap of propensity score distributions after trimming

3. Results

3.1 Classification models to predict tumor recurrence and long-term mortality

The evaluation of the classification models on the test set demonstrated good discriminative ability. For the recurrence prediction, between all the models, the backward selection one (*Table 1*) achieved the best results, ensuring the optimal balance between computational simplicity and predictive accuracy (AIC=514.657, BIC=539.2724, Cp=10.2273, time=5.8914s, AUC=0.765, Figure 3 (a)). The calculation of the AUC through cross-validation resulted in a value of 0.78 ± 0.04. The consistency between the cross-validated AUC and the initial test-set performance confirmed the model's stability and its reliable generalizability to unseen data. The cut-off threshold for the confusion matrix was determined by evaluating four different values: the default threshold (0.5), the threshold corresponding to the proportion of negative samples in the dataset, the optimal cut-off calculated by maximizing Youden's J statistic, and an intermediate custom of 0.35. Through the custom one, considered the best, the model yielded a True Positive Rate (TPR) of 0.69 and a False Positive Rate (FPR) of 0.31.

Table 1. Coefficients, odds ratios and p-values of the variables selected by backward selection for recurrence prediction.

variable	coefficient	Odds ratio	p-value	2.5% -97.5% CI OR	
num_log_nodes	1.0365	2.8193	0.0000	2.4835	3.2005
cat__size_>50	0.8402	2.3168	0.0000	1.5584	3.4444
cat__hormon_1	-0.3095	0.7338	0.0982	0.5085	1.0590
cat__grade_bin_1	0.7376	2.0910	0.0000	1.5998	2.7329
cat__pgr_pos_1	-0.4048	0.6671	0.0007	0.5275	0.8436

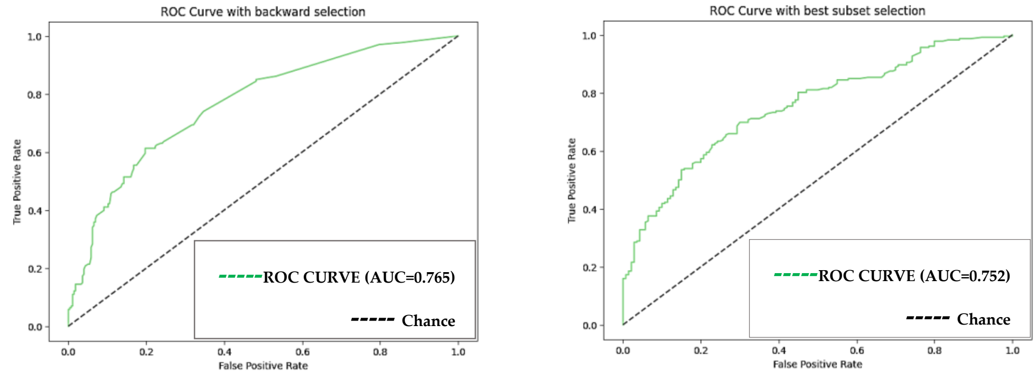
As detailed in *Table 1* the number of positive lymph nodes is the strongest predictor of recurrence. Similarly, large tumors and high-grade malignancy significantly increase the risk. These findings confirmed that morphological aggressiveness is a primary determinant of treatment failure. Progesterone receptor positivity acts as a significant protective factor, associated with a 33.3% reduction in the odds of recurrence. Hormonal therapy shows a borderline protective effect; this is likely because there are some limitations of standard models in isolating treatment effects within observational data.

For long-term mortality, the subset selection model (*Table 2*), was the optimal one, achieving an AUC (Figure 2 (b)) of 0.752 (CV AUC = 0.73). The optimal cut-off threshold (0.56) in the confusion matrix guaranteed a TPR of 0.70 and an FPR of 0.30.

Table 2. Coefficients, odds ratios and p-values of the variables selected by best subset selection for death prediction.

variable	coefficient	Odds ratio	p-value	2.5% -97.5% CI OR	
num__age	0.1716	1.1872	0.0095	1.0429	1.3516
num_log_nodes	0.7533	2.1239	0.0000	1.8299	2.4651
cat__size_<=20	-0.5315	0.5877	0.0000	0.4620	0.7476
cat__chemo_1	-0.3676	0.6924	0.0310	0.4958	0.9669
cat__grade_bin_1	0.5024	1.6527	0.0001	1.2754	2.1415

As shown in Table 2 the number of positive lymph nodes remains the most influential risk. High histological grade is associated with a 65% increase in the odds of death. Age shows a significant positive correlation with mortality, as initially hypothesized, while small tumors demonstrate a strong protective effect, reducing the odds of mortality by approximately 41% compared to larger tumor categories. Chemotherapy appears as a significant protective factor, suggesting a 31% reduction in the odds of death.

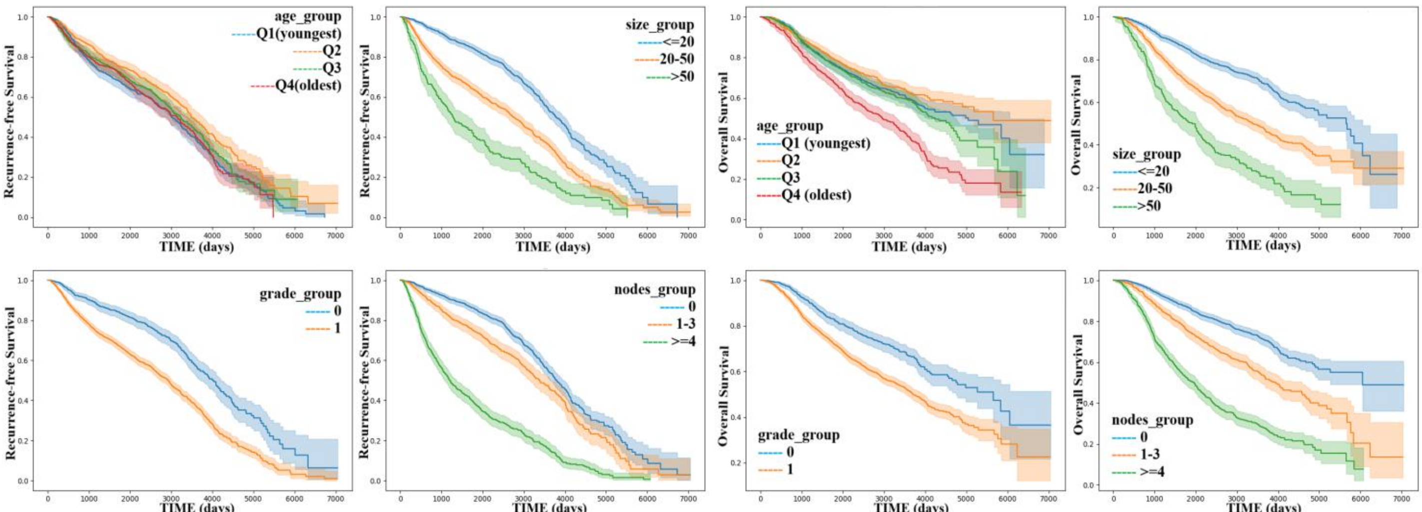


(a) (b)

Figure 3. ROC curve: (a) recurrence classification model (AUC=0.765); (b) death classification model (AUC=0.752)

3.2 Time-to-event models to examine recurrence-free and overall survival

Kaplan-Meier estimates highlighted significant differences in survival trajectories among clinical subgroups, as illustrated in the figure below (*Figure 4*). These visual distances were statistically confirmed by Log-Rank tests, which rejected the null hypothesis of equality between the survival distributions. To further quantify the temporal progression of the disease, the median survival time was calculated. The overall median recurrence-free survival was 3229 days, and the median overall survival was 4033 days. The median survival time was then calculated separately for treated and untreated patients.



(a) (b)

Figure 4. Kaplan-Meier curves: (a) on the left recurrence-free survival; (b) on the right overall survival.

In both analyses (recurrence-free and overall), the untreated group shows high medians (3442 days for recurrence and 4159 days for death) compared to those who received only hormonal therapy (2134 days for recurrence and 2591 days for death). This paradoxical outcome is a clear manifestation of confounding by indication.

Regarding recurrence-free survival, the semi-parametric analysis using Cox Proportional Hazards models yielded a Concordance Index (C-index) of 0.73 across all evaluated configurations (Full, Reduced, Feature Selection, and Regularized). Given this equivalent predictive performance, the reduced model was identified as the optimal choice due to its lower Akaike Information Criterion (AIC = 20161.08) and reduced model complexity. The analysis of Schoenfeld residuals revealed that some variables (log_nodes, size_<=20, hormon_1, and pgr_pos_1) did not meet the proportional hazards assumption over time. The model structure was then modified by applying stratification on *pgr_pos_1* and *hormon_1* (*Table 3*). This approach was not sufficient to resolve the non-proportionality of *log_nodes* and *size_<=20* , but this partial violation of the assumption was accepted to preserve the predictive power, acknowledging that the estimated hazard ratios for the non-proportional variables represent an average hazard effect over the entire follow-up period.

Table 3. Reduced Cox Stratified model's coefficients, hazard ratios and p-values. Time-to-event model to examine recurrence-free survival.

variable	coefficient	Hazard ratio	p-value	2.5% -97.5% CI HR	
age	-0.01	0.99	<0.005	0.98	0.99
log_nodes	0.68	1.97	<0.005	1.85	2.09
size_<=20	-0.30	0.74	<0.005	0.66	0.83
size_>50	0.25	1.29	<0.005	1.10	1.51
chemo_1	-0.49	0.61	<0.005	0.53	0.71
grade_bin_1	0.48	1.62	<0.005	1.42	1.85

In this model (*Table 3*) number of positive nodes remains the most critical factor. Both large tumours and high-grade tumours significantly accelerate the time to recurrence. Chemotherapy shows a strong protective effect in the survival context. This implies that, treated patients are 39% less likely to experience a recurrence than untreated patients. Small tumours decrease the hazard of recurrence by 26%, while the effect of the age is minimal (HR=0.99).

Regarding the analysis of overall survival, the reduced model was identified as the optimal choice (C-index = 0.71). Similarly to the previous analysis, the assessment of Schoenfeld residuals revealed that some variables (log_nodes, age and pgr_pos_1) did not meet the proportional hazards assumption over time. The model structure was modified by applying stratification on *pgr_pos_1 (Table 4*). This approach was not sufficient to resolve the non-proportionality of log_nodes and age, but this partial violation of the assumption was accepted to preserve the clinical interpretability of the hazard ratios and the deviation from proportionality is considered acceptable due to the large sample size.

Table 4. Reduced Stratified Cox model's coefficients, hazard ratios and p-values. Time-to-event model to examine overall survival.

variable	coefficient	Hazard ratio	p-value	2.5% -97.5% CI HR	
age	0.01	1.01	<0.005	1.01	1.02
log_nodes	0.58	1.79	<0.005	1.68	1.91
size_<=20	-0.32	0.73	<0.005	0.64	0.83
size_>50	0.30	1.35	<0.005	1.15	1.58
hormon_1	-0.35	0.71	<0.005	0.59	0.84
chemo_1	-0.22	0.81	0.01	0.68	0.95
grade_bin_1	0.30	1.36	<0.005	1.18	1.56

As shown in *Table 4* lymph nodes, larger tumours and high-grade tumours increase the risk of mortality by 79%, 35% and 36% respectively. Age indicates a small but significant increase in the risk of death for every additional year of age. Hormonal therapy emerges as a powerful protector reducing the hazard of death by 29%. Also chemotherapy shows a significant protective effect, associated with a 19% reduction in the hazard of mortality. Small tumors reduce the hazard of death by 27%, similarly to the recurrence case.

3.3 Causal inference analysis to estimate the effect of adjuvant hormonal therapy on recurrence and death

Initial observational models suggested a paradoxical effect, portraying hormonal therapy as a potential risk factor due to severe confounding by indication, as treatment was primarily administered to patients with worse baseline prognoses

(*Figure 1*). This selection bias was evident in the naive Average Treatment Effect (ATE), which incorrectly estimated a 2.27% (0.02265) risk increase for tumour recurrence and a 3.65% (0.03652) risk increase for long-term mortality among treated patients. However, after trimming the data to respect the propensity score positivity assumption and adjusting for selection bias through IPTW, the analysis revealed the true clinical benefit of the adjuvant therapy. In the ER+ cohort, the causal inference analysis estimated a weighted ATE of -0.1478 for tumour recurrence and -0.2357 for long-term mortality (risk reductions of 15% and 24%, respectively). The contrast between the naive ATE and the weighted ATE proves how critical it is to account for treatment assignment mechanisms in observational data. This effect is also highlighted by comparing the Odds Ratios (OR) of the hormonal therapy variable between the standard predictive models and the causal models. For tumour recurrence, the standard classification model yielded an OR of 0.7338 (26.62% reduction in odds), which underestimated the treatment's protective benefit compared to the causal model's OR of 0.5456 (45.44% reduction in odds). The discrepancy is bigger in long-term mortality. The standard full classification model considered the treatment as harmful (OR = 1.5649, 56.49% increase in odds), and the optimal best subset model failed to retain the variable as a significant predictor. In contrast, the causal framework correctly identified hormonal therapy as highly protective against mortality, yielding an OR of 0.3697 (63% reduction in odds).
These findings demonstrate that standard predictive models can distort or miss treatment efficacy due to confounding by indication, underscoring the necessity of causal inference frameworks when evaluating clinical observations.

4. Discussion and conclusion

In standard classification and semi-parametric survival models, tumour's extent and aggressiveness emerged as the primary drivers of adverse outcomes. Specifically, the number of positive lymph nodes (log_nodes) acted as the strongest predictor; this indicates that an increase in nodal burden increases both the odds and the hazard of recurrence and death. Similarly, the higher grade of the tumor roughly doubled the odds of recurrence and significantly increased the odds of long-term mortality. These findings are in accordance with previous studies [2,3]. Conversely, patients with smaller tumors (size_<=20) had significant protective effect on overall and recurrence-free survival. From a public health perspective, the strong protective effect of small tumour size underscores the vital importance of early detection programs and breast cancer screening. A central theme of this research was the role of confounding by indication. Initial descriptive analyses identified hormonal therapy as a risk factor, with treated patients reaching median recurrence faster than the untreated group. As hypothesized, this paradox arose because of confounding by indication. By balancing the cohorts based on their propensity score the true protective benefit of hormonal therapy was revealed. These findings are highly consistent with the results of the landmark EBCTCG meta-analysis [1] (RR_recurrence=0.68, RR_death=0.66), in fact both studies confirm that hormonal therapy significantly reduces the risk of adverse outcomes. Despite the robust findings, some limitations must be acknowledged. To strictly satisfy the positivity assumption required for IPTW, the dataset was trimmed by excluding specific subgroups (e.g., patients underwent to chemo). While this ensured a valid comparison between treated and untreated cohorts, it also reduced the sample size. Additionally, in the survival analysis, the Schoenfeld residuals test indicated that some key variables did not fully respect the proportional hazards assumption. Although it was accepted, this remains a limitation of the semi-parametric approach. Finally, the exclusion of non-clinical variables could result in residual confounding the propensity score might not have fully captured. This study successfully achieved its research objectives by identifying key prognostic drivers and quantifying treatment effects within an observational clinical setting. It was confirmed that lymph node involvement and histological grade are the most critical baseline determinants for both recurrence and long-term survival; it was demonstrated that reduced models, through feature selection, achieve optimal predictive accuracy, due to the exclusion of non-significant or redundant variable**s. T**he analysis also highlighted that standard predictive models can be misleading when used to evaluate treatment efficacy and only through the application of IPTW robust evidence was provided.

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